

Current Biology

Rapid homotopic communication between human orbitofrontal subregions

Highlights

- The orbitofrontal cortex (OFC) is a heterogeneous prefrontal cortical region
- It is unknown whether OFC exhibits homotopy comparable to lower cortical areas
- OFC stimulation in humans evoked homotopic contralateral responses
- Stimulation latencies were consistent with direct interhemispheric conduction

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In brief

Using intracranial cortico-cortical evoked-potential mapping in humans, Starkweather et al. reveal rapid, homotopic interhemispheric communication between bilateral orbitofrontal cortices (OFCs): medial OFC stimulation elicits responses in contralateral medial OFC, and lateral OFC stimulation elicits responses in contralateral lateral OFC. These responses have latencies consistent with direct interhemispheric conduction.

Report

Rapid homotopic communication between human orbitofrontal subregions

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SUMMARY

The orbitofrontal cortex (OFC) is a nexus for decision-making computations,¹ emotion,² and pain processing,³ with dissociable roles for its subfields: medial and anterior OFC are thought to represent economic value,^{4–6} whereas lateral OFC receives predominantly ipsilateral sensory information^{7,8} supporting stimulus-outcome associations.⁹ In areas such as primary sensorimotor cortices, homotopic interhemispheric coupling is a hallmark of cortical organization,¹⁰ but homotopy is weaker in the human prefrontal cortex,^{11,12} including the OFC, where anatomy is notably heterogeneous.^{13,14} Structurally, prefrontal commissural fibers course through the rostrum and genu of the corpus callosum^{15,16} and the anterior commissure,¹⁷ providing a substrate for interhemispheric communication in humans. Although macaque studies have demonstrated homotopy between select OFC subregions,^{18,19} whether OFC subregions communicate homotopically in humans remains unknown. Here, we use bilateral intracranial cortico-cortical evoked-potential (CCEP) mapping to show short-latency, homotopic interhemispheric connectivity across the medial/lateral axis of human OFC: the stimulation of medial OFC elicited rapid responses in contralateral medial OFC, whereas the stimulation of lateral OFC elicited rapid responses in contralateral lateral OFC. By demonstrating homotopic organization in an evolutionarily expanded, anatomically variable prefrontal territory, our data extend a canonical principle of cortical wiring into human OFC.

RESULTS AND DISCUSSION

We recorded from 156 orbitofrontal cortex (OFC) electrodes in seven participants implanted with bilateral OFC depth electrodes as part of presurgical evaluation for medication-refractory epilepsy, depression, or chronic pain (Table S1; Figure S1). Electrode arrays traversed the OFC from medial to lateral, allowing us to simultaneously capture activity across the medial/lateral axis of the OFC in both hemispheres (Figures 1A and 1B). To localize electrodes, we first manually identified constant OFC sulci based on cortical surface reconstruction (Figure 1C). We localized each electrode relative to these constant sulci: olfactory sulcus (OlfS), medial orbital sulcus (MOS), lateral orbital sulcus (LOS), and transverse orbital sulcus (TOS). The anatomical subregion of each electrode is defined in relation to these sulci (Figure 1A). For example, all electrodes lateral to the LOS are classified as residing in the lateral OFC (Figure 1D). Furthermore, sulci provided a consistent anatomical coordinate system, following the electrode localization framework from Rich and Wallis.²⁰ Using

the LOS as a reference point, we calculated the medial/lateral distance of each electrode from the LOS to assign its medial/lateral coordinate along the OFC surface.

We performed a cortico-cortical evoked-potential (CCEP) mapping protocol (STAR Methods) and found that OFC stimulation in one hemisphere produced a rapid evoked response in the contralateral hemisphere with an average latency of 25 ms. Strikingly, the medial/lateral coordinate of OFC stimulation determined the location of the response in the contralateral OFC. Stimulation of the medial OFC produced CCEPs with large N1 deflections in more medial contralateral OFC electrodes (Figure 2A), while stimulation of the lateral OFC produced large N1 deflections in more lateral contralateral OFC electrodes (Figure 2B). We used the standard method of employing a threshold of 5 Z scores within 10–50 ms following stimulation to define a significant N1 response to stimulation.^{21,22} We chose this narrow time window to capture short-latency direct responses (“N1”) rather than more delayed responses reflective of local current spread.^{23,24}

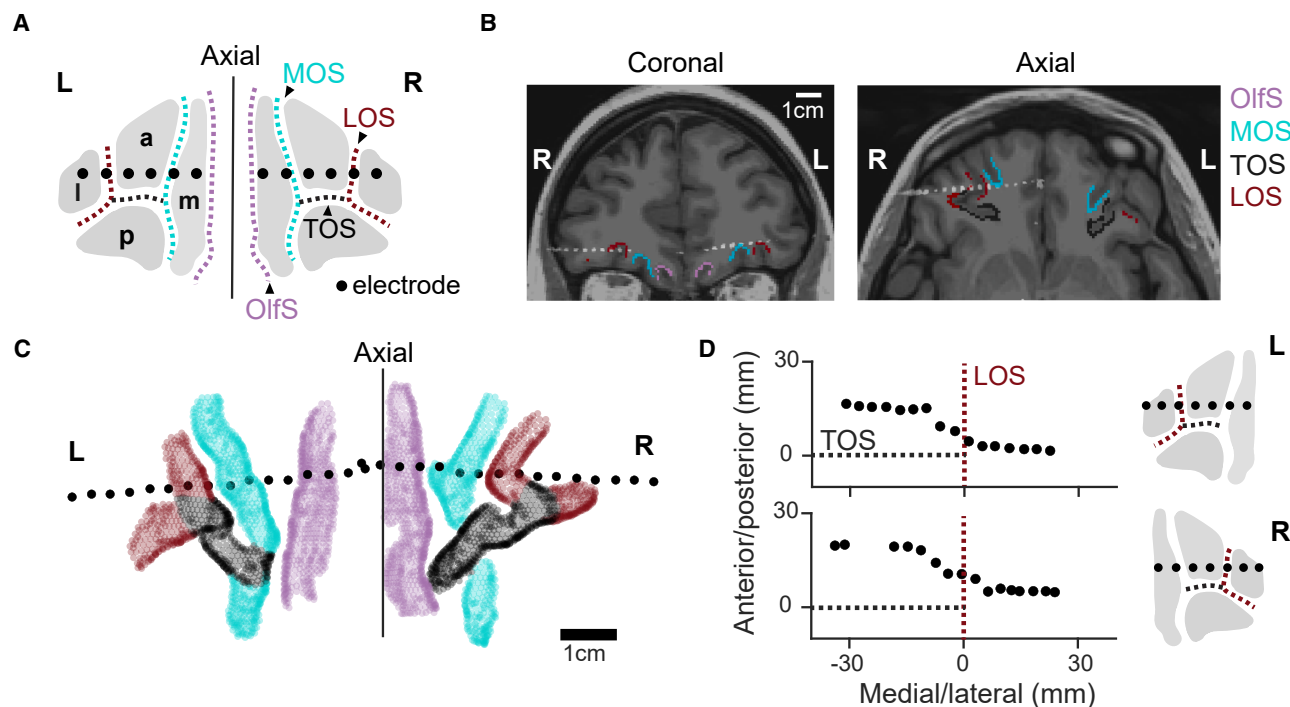


Figure 1. Electrode localization and coordinate system

(A) Schematic of orbitofrontal cortex (OFC) subregions and constant sulci. Medial OFC (m), anterior OFC (a), posterior OFC (p), and lateral OFC (l) are shown. (B) Post-operative computed tomography (CT) overlaid on T1-weighted magnetic resonance imaging (MRI) to demonstrate electrode placement in an example participant. Only electrodes within the frame of the displayed 1 mm coronal/axial MRI slices are visible as radioopaque bright spots. Manually labeled sulci are shown in color.

(C) Sulci plotted in the axial plane alongside all electrodes in the same example participant.

(D) Electrodes from the same example participant transformed into "sulcal space," defined in the medial/lateral dimension by the LOS.

See also Figure S1.

Anatomical symmetry in contralateral evoked N1 responses existed over the population of recorded electrodes across all participants. We plotted a heatmap displaying the probability of an evoked N1 over the contralateral medial/lateral OFC axis versus the ipsilateral OFC stimulation site (Figure 3A). The distribution of significant responses preferentially lay along the diagonal of the heatmap, reflecting responsive contralateral sites that mirrored the ipsilateral stimulation site. This pattern was preserved at lower stimulation intensities, with similar response latencies but fewer overall significant N1s (Figures 4A–4C). This distribution was different than expected based on the null distribution ($\chi^2(437) = 601.26$, $p = 2.81 \times 10^{-7}$). To confirm that this distribution was consistent across individual subjects, we computed a "mirror index" (M.I.) for each subject:

$$M.I. = \text{mean}(|C_{null} - I_{null}|) - \text{mean}(|C_{N1} - I_{N1}|), \quad (\text{Equation 1})$$

where C is the medial/lateral coordinate of the contralateral recording site and I is the medial/lateral coordinate of the ipsilateral stimulation site. C_{N1} and I_{N1} refer to coordinates with evoked N1s, and C_{null} and I_{null} refer to all coordinates stimulated and recorded. If C_{N1} and I_{N1} tended to occur in more symmetric locations on opposing hemispheres compared with C_{null} and I_{null} , M.I. would be positive (Figure 3B). All subjects with significant evoked responses (6 of 7 subjects) exhibited a positive M.I. (M.I. versus zero: $t(5) = 5.60$, $p = 0.0025$, $SD = 1.72$). There

was no difference in M.I. depending on whether the left or right hemisphere was stimulated (medians: 4.39 versus 3.99; Wilcoxon rank-sum, $W = 29$, $p = 0.84$; Figures 4D–4F), nor was there a difference in M.I. between subjects implanted for refractory epilepsy versus other conditions (medians: 4.27 versus 2.84; Wilcoxon rank-sum, $W = 10$, $p = 1.00$; Figure S2). M.I. values in the two female subjects were within the range of values observed in male subjects. In summary, the anatomical location of evoked contralateral responses mirrored the ipsilateral stimulation site.

We next asked whether specific OFC subregions served as hotspots of interhemispheric OFC communication. We found that stimulation of the medial OFC evoked an N1 in 57% of recording sites in the contralateral medial OFC (Figure 3C). Stimulated and recorded contralateral medial OFC electrodes were outside the estimated radius of activated tissue (STAR Methods; Figure S3). Even the closest ipsilateral-contralateral medial OFC electrode pairs were separated by ≥ 2 cm across the midline, making direct current spread an unlikely explanation for the high proportion of evoked responses. Stimulation of the ipsilateral medial OFC also evoked a significant N1 in 29% of recording sites in the contralateral anterior OFC. The next highest proportion of symmetrically connected subregions occurred between the two lateral OFCs, with ipsilateral lateral OFC stimulation evoking a significant N1 in 27% of contralateral lateral OFC

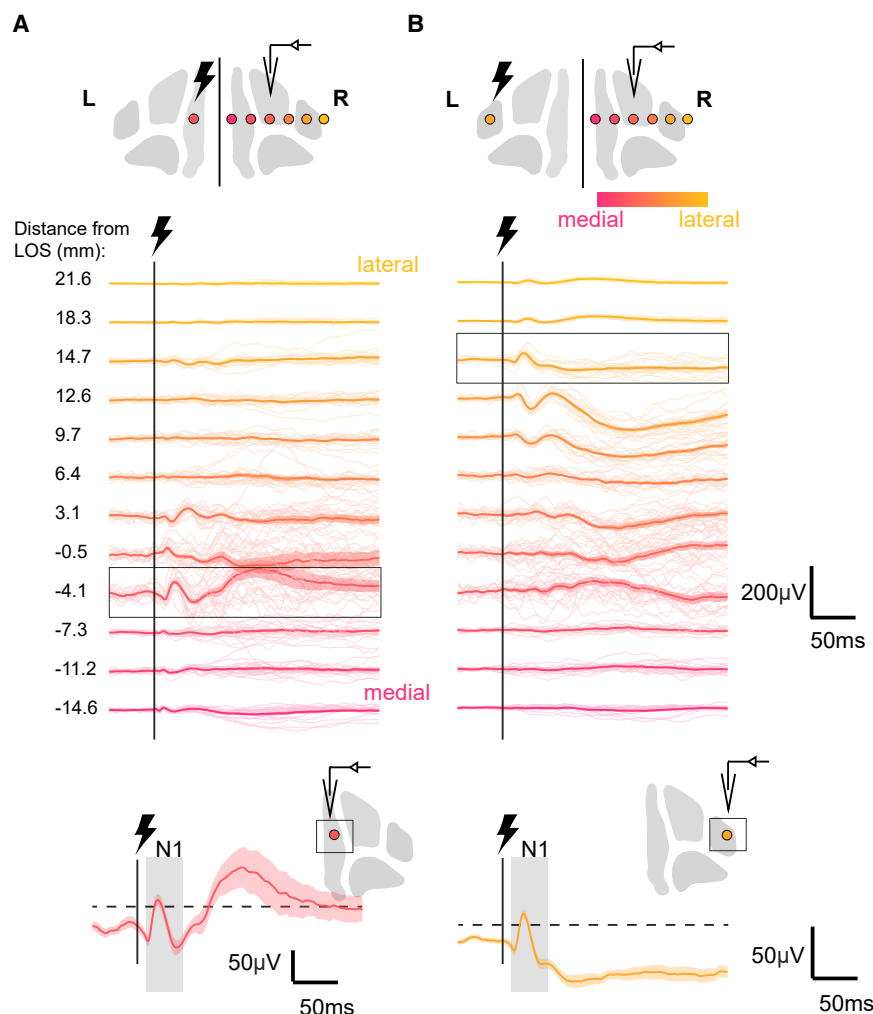


Figure 2. CCEPs in the contralateral OFC are anatomically constrained by the medial/lateral location of OFC stimulation

(A) In this example patient (patient #3, implanted for refractory depression), stimulation within the left-sided medial OFC evokes CCEPs in the right-sided medial/anterior OFC channels. The inset of the OFC channel in the box demonstrates the time window for capturing the N1 response (10–50 ms post-stimulation), and the dotted line indicates a threshold of 5 Z scores above baseline. (B) Stimulation within the left-sided lateral OFC evokes CCEPs in the right-sided lateral OFC channels. Mean $\pm$ SEM traces are plotted.

multiple intermediate sulci between the MOS and LOS, while others lack them entirely.¹⁴ Despite this variability, and despite the general decline in homotopic organization from Rolandic regions to prefrontal/association cortices,^{11,12} the principle of homotopy appears preserved within the OFC.

Homotopic connections provide a mechanism for efficient decision-making despite lateralized sensory representations. The OFC is unique among prefrontal cortical regions in that it receives input from all five sensory modalities.⁷ These sensory inputs converge in the lateral OFC, where they are thought to generate stimulus-outcome associations, whereas the medial and medial/anterior OFC encode economic value and mediate value comparison.⁹ Consider a case where two stimuli are displayed in opposite visual hemifields, and the left hemifield's stimulus carries greater value. Even without eye movements, OFC preferentially encodes the more valuable option, reflecting covert attention.²⁸ Visual input from each hemifield is registered in the contralateral OFC. Thus, the more valuable left-sided stimulus would first arrive in the right lateral OFC and ultimately be converted into a value representation in the right medial/anterior OFC. To enable comparison with the right-sided stimulus, information from both hemifields must converge in the medial/anterior OFC. This could occur either by transferring sensory information across hemispheres or by transmitting value signals directly. Our findings demonstrate the existence of a rapid, direct interhemispheric route supporting both possibilities, thereby providing a biologically efficient substrate for integrating lateralized sensory inputs into unified value-based decisions.

electrodes (Figure 3C). Stimulation in the lateral OFC evoked an N1 in only 11% of contralateral anterior OFC electrodes. The average latency of the evoked N1 was 25.17 ms (SD 8.30 ms) between ipsi- and contralateral medial OFCs and 25.57 ms (SD 9.20 ms) between ipsi- and contralateral lateral OFCs (Figure 3D). A two-sample Kolmogorov-Smirnov test indicated no significant difference between the two latency distributions ($D = 0.171$, $p = 0.709$), reflecting similarly rapid responses regardless of the stimulated subregion.

The bilateral OFCs have anatomical connectivity through the genu and rostrum of the corpus callosum^{15,16,25} and additional connectivity through the anterior commissure,¹⁷ spanning an estimated inter-OFC path of ~ 10 cm. Given an estimated conduction speed of 4–8 m/s,^{23,26} the ~ 25 ms latency between ipsilateral and contralateral medial OFCs is compatible with rapid, direct interhemispheric signaling. Importantly, because our CCEP data do not resolve individual white-matter tracts, we cannot ascribe the observed responses to a specific commissural pathway, and they may reflect transmission through the corpus callosum, the anterior commissure, or both.

The OFC is among the evolutionarily youngest parts of the brain,²⁷ notable for substantial interindividual anatomic variability¹³ and asymmetry.¹⁴ For example, some individuals exhibit

multiple intermediate sulci between the MOS and LOS, while others lack them entirely.¹⁴ Despite this variability, and despite the general decline in homotopic organization from Rolandic regions to prefrontal/association cortices,^{11,12} the principle of homotopy appears preserved within the OFC.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to be fulfilled by the lead contact, Clara Starkweather (clara.starkweather@ucsf.edu).

Materials availability

This study did not generate any new, unique reagents.

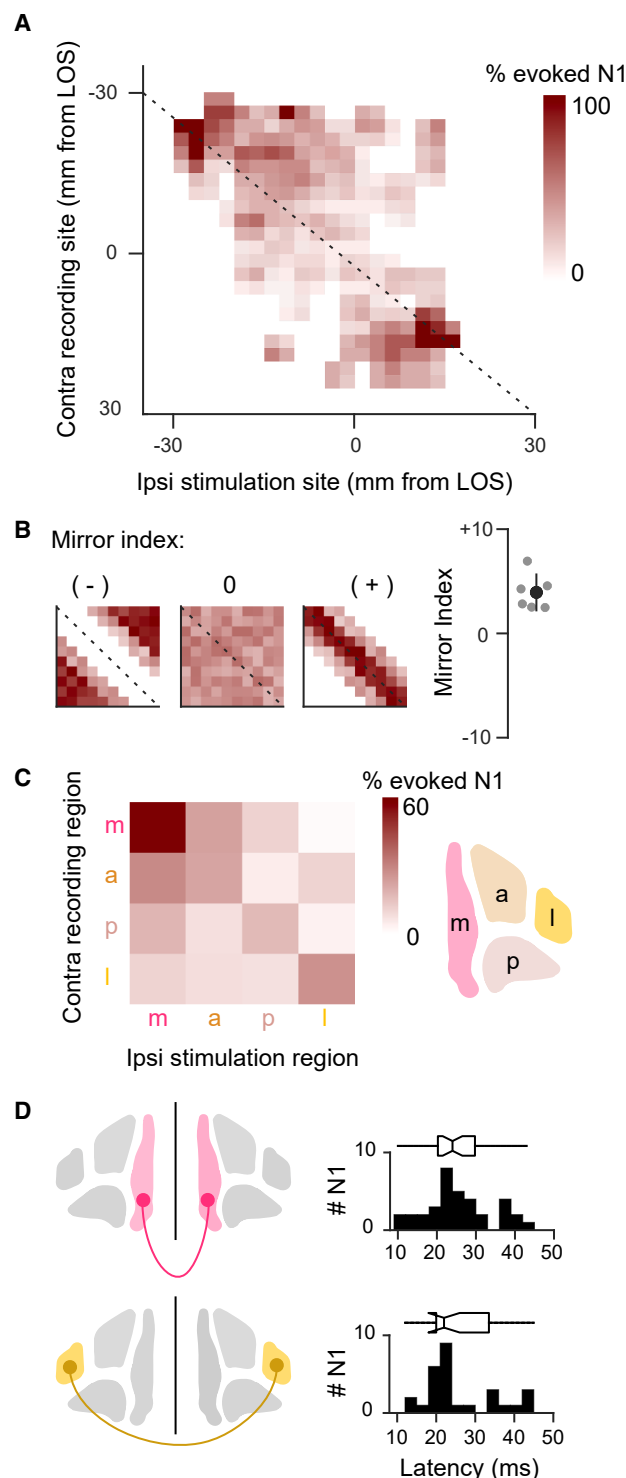


Figure 3. Rapid, homotopic communication between bilateral OFCs across participants

(A) Heatmap displaying probability of evoked N1 by medial/lateral coordinate of the stimulation site versus the medial/lateral coordinate of the contralateral recording site.

(B) M.I. >0 indicates a tendency toward homotopic connectivity, shown in all subjects with significant N1s. Mean $\pm$ SD shown.

(C) Subregion analysis, in which ipsilateral stimulation and contralateral recording sites were divided into medial OFC (m), anterior OFC (a), posterior

Data and code availability

- All data and code presented in the main results are available through Figshare at <https://doi.org/10.6084/m9.figshare.31083118>.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

C.K.S., E.F.C., and R.T.K. conceived the study and established the methodology. C.K.S., K.S.W., and E.H.W. visualized the data. E.F.C., R.T.K., and A.D.K. acquired the funding. E.F.C., R.T.K., K.K.S., J.T.W., P.B., P.H., and P.S. collected the data. E.F.C. and R.T.K. supervised the project. C.K.S. and R.T.K. wrote the manuscript. C.K.S., P.S., R.T.K., E.F.C., K.S.W., and E.H.W. reviewed and edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2026.01.076>.

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OFC (p), and lateral OFC (l). mOFC and IOFC had the greatest tendency to evoke CCEPs in their contralateral counterpart.

(D) Interhemispheric latency of $\sim$ 25 ms in both mOFC and IOFC homotopic connections. Bottom and top edges of boxes denote 25th and 75th percentiles. See also [Figure S2](#).

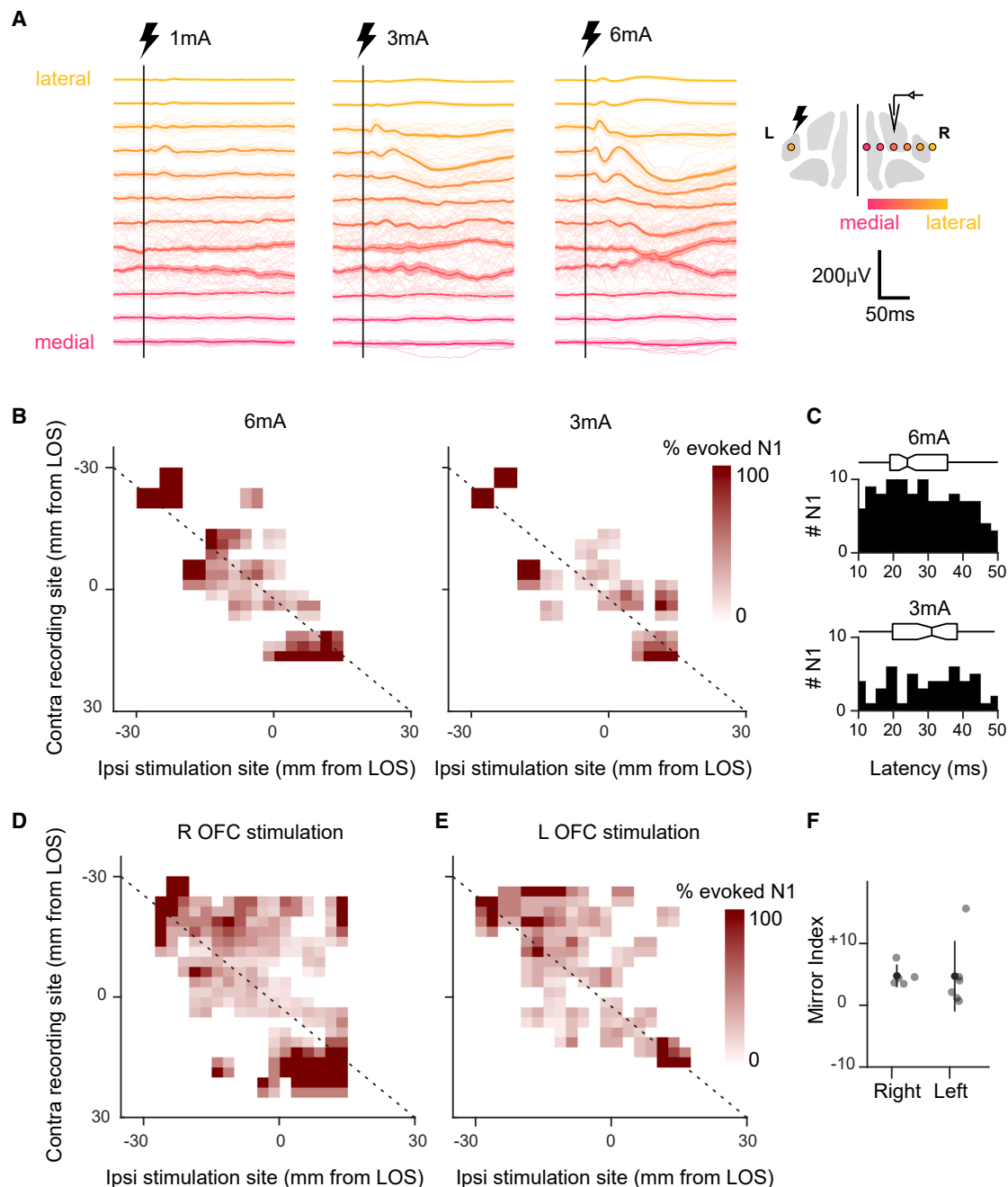


Figure 4. Homotopic connectivity demonstrated over stimulation intensities and separate hemispheres

(A) Single subject example (same subject as in Figure 2), demonstrating responses over right OFC electrodes with left lateral OFC stim at 1, 3, and 6 mA.
 (B) Heatmap displaying the probability of evoked N1 by medial/lateral coordinate of the ipsilateral site versus the medial/lateral coordinate of the contralateral stimulation site, for 4 subjects in whom stimulation protocols were conducted at 3 and 6 mA.
 (C) CCEP onset latencies did not differ significantly between N1 responses evoked at 6 mA ($n = 116$; median = 24 ms; mean = 26.9 ms) and 3 mA stimulation ($n = 49$; median = 31 ms; mean = 29.5 ms; Wilcoxon rank-sum test, $p = 0.16$).
 (D) Heatmap displaying the probability of evoked N1 by the medial/lateral coordinate of the right hemispheric stimulation site versus the medial/lateral coordinate of the left-sided recording site.
 (E) Heatmap displaying the probability of evoked N1 by the medial/lateral coordinate of the left hemispheric stimulation site versus the medial/lateral coordinate of the right-sided recording site.
 (F) M.I. was not significantly different depending on whether the left or right hemisphere was stimulated. Medians: 4.39 (right OFC stimulated) versus 3.99 (left OFC stimulated); Wilcoxon rank-sum, $W = 29$, $p = 0.84$. Mean $\pm$ SD shown.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Deposited electrophysiology data	This paper	10.6084/m9.figshare.31083118
Software and algorithms		
MATLAB 2023a	Mathworks	RRID: SCR:001622
Custom MATLAB codes	This paper	10.6084/m9.figshare.31083118

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Data were obtained from 7 patients (Table S1) who had stereotactically implanted intracranial depth electrodes (AD-Tech, PMT, or Dixi, 3–5mm inter-electrode spacing) placed at the University of California, San Francisco (UCSF, n = 6 subjects) or at Washington University in St. Louis (WUSTL, n = 1 subject). 3 patients were implanted as part of a clinical trial for refractory depression (NCT04004169) and 1 patient was implanted as part of a clinical trial for refractory chronic pain (NCT04144972, Table S1). Electrode placement was exclusively guided by clinical or parent trial research needs. Approval for the study was granted by the institutional review boards of the UCSF, UC Berkeley and WUSTL. Written informed consent was obtained from all subjects prior to testing.

METHOD DETAILS

Cortico-cortical evoked potential mapping protocol

We recorded from a total of 156 electrode contacts located in the OFC. Intracranial electroencephalography data were acquired using a multichannel pre-amplifier connected to a digital signal processor and digitized at 3–10kHz using either the Tucker-Davis Technologies or Nihon Kohden recording system. There were no seizures recorded during any epochs. We mapped cortico-cortical evoked potentials (CCEPs) by performing 15–30 pulses of either 0.5Hz or 1Hz bipolar stimulation with a pulse-width of 100us. Adjacent contacts were chosen as the bipolar anodic and cathodic contact. For example, stimulation of contact 5 means that bipolar stimulation between contact 5 and contact 6 was applied. Stimulation was applied between 1mA and 10mA. The maximum stimulation amplitude was determined by individualized clinical need, based on previously safety tested parameters. All results presented in the text were chosen from highest possible amplitude applied to that subject, which ranged from 6–10mA.

QUANTIFICATION AND STATISTICAL ANALYSIS

Imaging analysis and electrode localization

Each structural T1-weighted magnetic resonance imaging (MRI) scan was first processed through FreeSurfer v6.0.0 to create 3D pial and inflated cortical reconstructions.^{29,30} Next, all sulci were defined on these cortical reconstructions using the FreeSurfer curvature metric to differentiate between sulcal and gyral components.^{29–31} Trained neuroanatomists (E.H.W. and K.S.W.) manually identified OFC sulci in each hemisphere (N = 7; 14 hemispheres) using FreeSurfer's tksurfer tools and guided by the most up-to-date definitions of OFC sulcal organization.^{14,32} The OFC sulci of interest in the present study were as follows: (1) olfactory sulcus (OlfS), (2) transverse olfactory sulcus (TOlfS), (3) transverse orbital sulcus (TOS), (4) anterior section of the medial orbital sulcus (MOS-a), (5) posterior section of the medial orbital sulcus (MOS-p), (6) anterior section of the lateral orbital sulcus (LOS-a), (7) posterior section of the lateral orbital sulcus (LOS-p), (8) intermediate orbital sulcus (IOS), (9) posterior orbital sulcus (POS), and (10) sulcus fragmentosus (SF). Prior research has shown that the OlfS, TOlfS, TOS, MOS-a, MOS-p, LOS-a, and LOS-p are present in all individuals, whereas the IOS, POS, and SF are more variably present, consisting of a number of components ranging from 0–4.^{13,14} The seven consistent sulci were present in all patients. Example sulcal labels for constant sulci MOS (includes MOS-a and MOS-p), LOS (includes LOS-a and LOS-p), TOS, and OlfS are shown in Figure 1.

Electrodes were localized based on co-registration of 1mm slice thickness T1 weighted MRI and 1mm slice thickness bone window computed tomography (CT) scans. Following co-registration in FreeSurfer, electrode locations were marked using the region of interest (ROI) tool. The centroid coordinates, as well as coordinates from manually defined OFC sulcal labels, were exported in patient space for further analysis in MATLAB (Mathworks, Natick, Massachusetts). We created a standardized coordinate system across subjects based on the constant sulci MOS (combining MOS-a and MOS-p) and TOS²⁰ (Figure 1D). The anterior/posterior coordinate

was measured in relation to the TOS. For each electrode, we calculated the distance between the anterior/posterior electrode coordinate and the anterior/posterior coordinate of the centroid of the points contained in the TOS label that lay within the same medial/lateral dimension as the electrode. If the electrode lay beyond the medial/lateral dimension of the TOS, we measured the anterior/posterior distance from the closest TOS medial/lateral slice containing the TOS. The medial/lateral coordinate was measured in relation to the LOS. For each electrode, we calculated the distance between the medial/lateral electrode coordinate and the medial/lateral coordinate of the centroid of the points contained in the LOS label that lay within the same anterior/posterior dimension as the electrode. For anatomic localization, electrodes were excluded if they lay more than 1mm medial to the centroid of the olfactory sulcus (placing the electrode diameter fully within the gyrus rectus); electrodes were considered to be in the medial OFC if they were otherwise medial to MOS; in the anterior OFC if they were lateral to the MOS, anterior to the TOS, and medial to the LOS; in the posterior OFC if they were lateral to the MOS, posterior to the TOS, and medial to the LOS; and in the lateral OFC if they were lateral to the LOS. Electrodes in the inferior frontal gyrus were excluded. Electrodes in the gyrus rectus were excluded because stimulation and recording were available in only one subject, and in that case the sole stimulation intensity tested (6 mA) produced an estimated radius of activated tissue that overlapped the contralateral gyrus rectus electrodes.

Data preprocessing

Epilepsy patients' data was manually examined in subsecond epochs during CCEP recordings to confirm the absence of epileptiform activity. Interictal discharges were excluded. Furthermore, the OFC was not identified as an epileptiform locus in any of the patients. Data were visually inspected and flat or noisy channels were removed. The fieldtrip package in MATLAB was used for the following pre-processing steps. First, continuous data were band-pass filtered between 0.5 and 200 Hz using a second-order Butterworth filter (two-pass, zero-phase). Next, we applied 10th-order Butterworth IIR band-stop filters (two-pass, zero-phase) centered at 60 Hz and its harmonics (120, 180, 240, 300 Hz). The notch filter bandwidth was adjusted depending on the subject (0.2 Hz for most subjects, and 0.5 Hz for patients #2 and #6 based on visual inspection of the power spectra). Filtering was implemented in MATLAB using the `designfilt` and `filtfilt` functions. Finally, bipolar referencing was applied using the nearest neighboring electrode. Electrodes were only included in the dataset if a nearest neighbor for bipolar referencing was available. Following elimination of noisy channels and the exclusion of redundant electrodes introduced by bipolar referencing, 124 electrodes were included in the final dataset.

Modeling volume of tissue activated

All results presented in the text were the highest amplitude applied to that subject, which ranged from 6–10mA. We used a spherical volume of tissue activated model derived in Dembek et al. to model the radius of tissue activated using the following equation^{33,34}:

$$r = \left(\frac{pw}{90 \mu s} \right)^{0.3} * \text{sqr}t \left(0.72 \frac{Vm}{A} * \left(\frac{I}{165 \frac{V}{m}} \right) \right) \quad (\text{Equation 2})$$

Where *pw* was our pulse-width of 100us and *I* was the applied current. See Dembek et al. for details and derivation of other constants. The estimated radius of tissue activated *r* at an applied stimulation current of 6–10mA (the settings plotted throughout the main figures) is approximately 5–7mm (Figure S3A). This should be considered an upper bound, as existing volume-of-tissue-activated models are based on monopolar stimulation and therefore overestimate spread relative to bipolar stimulation.

To characterize stimulation intensity at the electrode-tissue interface, we computed the interface current density for each stimulation amplitude as $J = I/A$, where *I* is the stimulation current and *A* is the contact surface area. We computed the contact surface area as a cylinder based on electrode dimensions (diameter range = 0.80–0.86mm, length = 2mm), indicating that current density ranged from 1.11–1.86mA/mm² at an applied stimulation current of 6–10 mA (Figure S3B).

Although the estimated radius of tissue activated overlapped both gray and white matter for all stimulation pairs, we nonetheless examined whether responses differed depending on whether the stimulating contacts were located within cortical gray matter. Specifically, we calculated the proportion of significant N1 responses when both anodic and cathodic electrodes were embedded within the same gray-matter ribbon. We found that 19.5% of contralaterally recorded contacts exhibited significant N1 responses when both ipsilateral contacts were in gray matter, compared with 17.2% when they were not. Thus, we observed no marked difference in response probability as a function of whether stimulation occurred in gray versus white matter, consistent with the expectation that the activated volume spans both tissue types.

Detection of N1 responses and latency estimation

For each stimulation site, we constructed peri-stimulus time histograms (PSTHs) from bipolar recordings aligned to the stimulation onset. Each trace was baseline-subtracted using the mean voltage value of the 200ms preceding the stimulus. Standard error was plotted around this mean on all PSTHs throughout the manuscript. For each recording channel, the mean evoked trace was z-scored relative to the baseline mean and standard deviation for that specific channel. A response was considered a significant N1 if the z-scored waveform exceeded a threshold of 5 within a 10–50ms post-stimulus window. The onset latency was defined as the first sample within this window at which the z-score crossed threshold. Latencies were referenced to the stimulus onset (0ms).